

URITECT Sex-Specific Bayesian UTI Research Specification v1.2

Female model: uti_bayesian_female_v1_1_20260926 **Male candidate model:** uti_bayesian_male_v0_1_20261001 **Status:** Implemented research models; literature-derived and not locally calibrated **Reference standard in principal studies:** Urine culture / microbiologically confirmed UTI

1. Intended Use

URITECT provides offline ten-analyte strip classification, safety assessment, renal follow-up, and a sex-specific Bayesian UTI research estimate for eligible symptomatic adults aged 18-64. Neither estimate diagnoses UTI, recommends antibiotics, replaces urine culture, or represents a locally calibrated probability.

Female and male calculations are separate. Female values are never applied to male users. The male model uses only nitrite and leukocyte esterase because no adequate male-specific blood LR was identified.

2. Eligibility

Both models require confirmed age 18-64, no urinary catheter, no known urinary tract abnormality, no immunocompromise, at least one qualifying acute urinary symptom, and no systemic finding that moves the user to the safety pathway.

Female calculations additionally require confirmed nonpregnancy. Dysuria, frequency, urgency, suprapubic pain, or visible hematuria can satisfy the female symptom gate. Vaginal discharge or irritation stops the calculation and activates alternate-cause consultation guidance.

Male calculations additionally require no diabetes and no suspected sexually transmitted infection, matching the principal male study exclusions. Dysuria, frequency, or urgency must be present. The study included men aged 18 years and older; use in URITECT's narrower 18-64 population remains a transportability limitation.

3. Female Model

The female v1.1 model is unchanged. It uses a provisional prior of 0.50, one mutually exclusive dipstick LR, and one mutually exclusive symptom LR.

Female dipstick pattern	LR
Nitrite positive plus source-threshold leukocytes or blood	7.20
Nitrite positive without the stronger grouped pattern	5.50
Blood positive when nitrite is not positive	1.70
Leukocytes positive when nitrite and blood are not positive	1.40
Nitrite, leukocytes, and blood all confirmed negative	0.22

Female symptom pattern	LR
Dysuria and urgency	1.50
Dysuria	1.30
Urgency	1.20
Frequency	1.10

See URITECT_Bayesian_UTI_Scoring_System_v1.1.md for the complete female threshold mapping, evidence, and limitations.

4. Male Evidence and Prior Derivation

Den Heijer et al. enrolled 603 symptomatic men from Dutch general practices. Complete data were available for 490; 321 had microbiologically confirmed UTI at at least 10^3 CFU/mL. The median age was 65 years (range 18-97). The study excluded relevant urological or nephrological comorbidity other than benign prostatic hypertrophy, diabetes, immunocompromising disease, catheterization, and suspected sexually transmitted infection.

The full cohort prevalence was $321/490 = 65.5\%$, but URITECT does not use that as its prior because the cohort was older than the app population. Table 3 reported 224 culture-positive and 80 culture-negative participants in its older-age category. Therefore, in the complementary younger subgroup:

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culture positive = 321 - 224 = 97
culture negative = 169 - 80 = 89
candidate prior = 97 / (97 + 89) = 0.521505... (52.2%)
candidate prior odds = 97 / 89 = 1.089888...

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This is a transparent subgroup derivation, not a published Filipino prevalence estimate. The article describes the age boundary as both ≥ 60 in narrative text and >60 in a table, so URITECT does not claim an exact birthday boundary for this derived subgroup. It does not exactly represent the app's 18-64 population and is frozen only as a provisional research prior pending local culture-confirmed calibration.

5. Male Ordered-Threshold Model

Den Heijer et al. assigned one point to positive leukocyte esterase and two points to positive nitrite. Table 2 reported the following thresholds:

Published threshold	LR+	LR-	95% CI for LR+ / LR-
Score at least 1: LE or nitrite positive	1.65	0.35	1.41-1.94 / 0.28-0.45
Score at least 2: nitrite positive	4.87	0.48	3.19-7.43 / 0.42-0.55
Score 3: nitrite and LE positive	5.14	0.54	3.24-8.17 / 0.48-0.60

URITECT applies the first matching row below and never multiplies these correlated thresholds:

Male rule	Pattern	Candidate LR
M1	Nitrite positive and LE positive	5.14
M2	Nitrite-positive threshold met without M1	4.87
M3	LE-positive threshold met without M1 or M2	1.65
M4	Nitrite and LE both confirmed negative	0.35
M5	Missing, unreliable, or unmatched	1.00; no calculation

M2 and M3 use an ordered-threshold approximation: the publication reports threshold LRs, not mutually exclusive exact-pattern LRs. This is an explicit URITECT design assumption requiring statistical review and prospective validation. It is not hidden as a published clinical rule.

The male study treated any LE color change as positive. URITECT therefore maps Trace 15, Small 70, Moderate 125, and Large 500 to male LE-positive. Nitrite requires Positive. Blood is not numerically weighted in the male UTI model.

6. Equation

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prior odds = prior probability / (1 - prior probability)
posterior odds = prior odds x the one selected LR
posterior probability = posterior odds / (1 + posterior odds)

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Female calculations multiply one dipstick LR and one symptom LR. Male calculations use one ordered dipstick-threshold LR only.

7. Safety and Display Rules

- Fever/chills, flank pain, significant nausea/vomiting, inability to maintain hydration or medication, severe weakness, confusion, or fainting remain outside the ordinary posterior and enter the safety pathway.
- Visible hematuria is consultation context and is not numerically weighted.

- No Low/Moderate/High probability bands or treatment thresholds are used.
- The app displays the sex-specific model version, starting prior, selected LR, and provisional posterior with a screening-only warning.
- Local records store calculation status, prior, posterior, factors, and model version.

8. Validation Boundary

The calculations are mathematically reproducible, but neither complete model is clinically calibrated for the local population. Expert content review can assess relevance and implementation clarity. Clinical validity requires physician and statistical review plus prospective culture-confirmed patient-level validation, including calibration, discrimination, confidence intervals, and subgroup performance.

9. References

1. den Heijer CDJ, van Dongen MCJM, Donker GA, Stobberingh EE. Diagnostic approach to urinary tract infections in male general practice patients: a national surveillance study. *Br J Gen Pract*. 2012;62:e780-e786. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3481519/>
2. Koeijers JJ, Kessels AGH, Nys S, et al. Evaluation of the nitrite and leukocyte esterase activity tests for the diagnosis of acute symptomatic urinary tract infection in men. *Clin Infect Dis*. 2007;45:894-896. <https://pubmed.ncbi.nlm.nih.gov/17806056/>
3. Kurotschka PK, Gagyor I, Ebell MH. Acute Uncomplicated UTIs in Adults: Rapid Evidence Review. *Am Fam Physician*. 2024;109:167-174.
4. Little P, Turner S, Rumsby K, et al. Developing clinical rules to predict urinary tract infection in primary care settings. *Br J Gen Pract*. 2006;56:606-612.
5. Bent S, Nallamothu BK, Simel DL, Fihn SD, Saint S. Does this woman have an acute uncomplicated urinary tract infection? *JAMA*. 2002;287:2701-2710.